

AI와 딥러닝을 활용한 경영분석

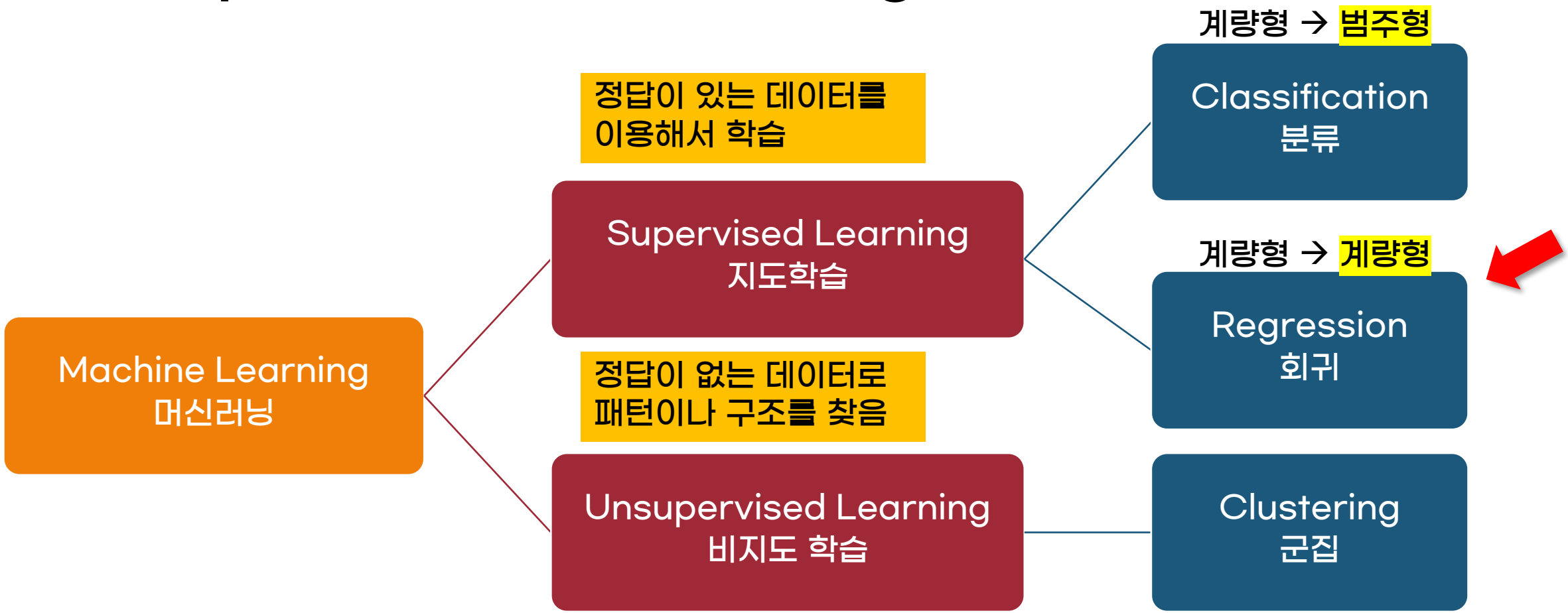
Week 5

서울과학기술대학교
SEOULTECH Seoul National University of Science & Technology

Topics

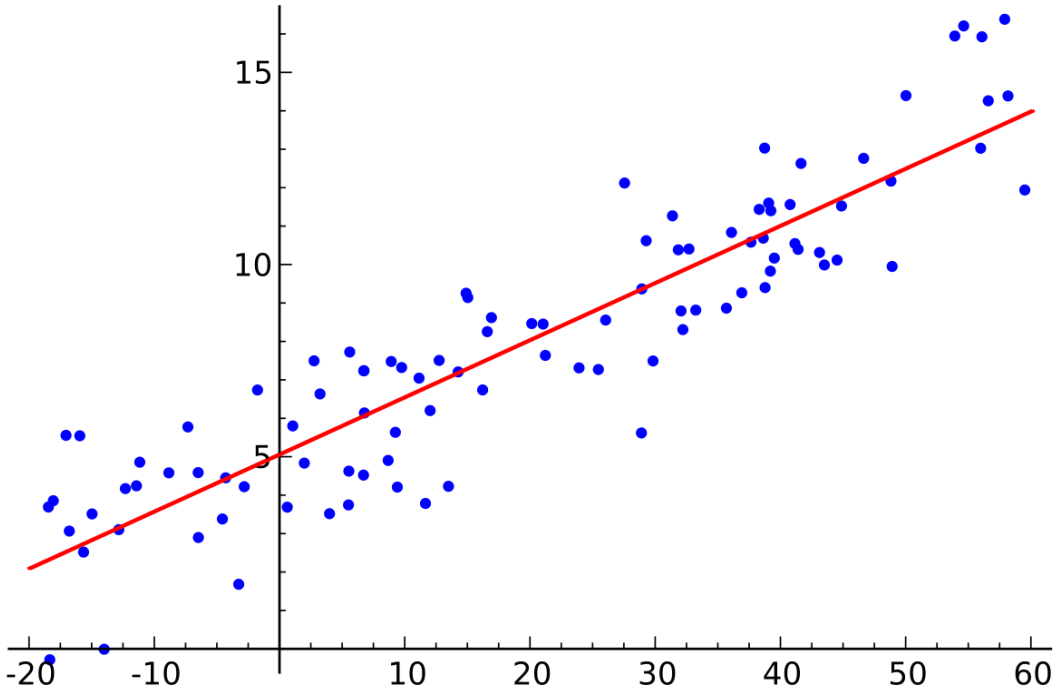
- 지도학습 I: 회귀 분석
- 매출 예측 실습

지도학습과 비지도학습 (Supervised and Unsupervised Learning)



Regression 회귀분석

- In statistical modeling, regression analysis is a set of statistical processes for estimating the relationships among variables. 변수들 사이의 관계를 추정하는 통계기법



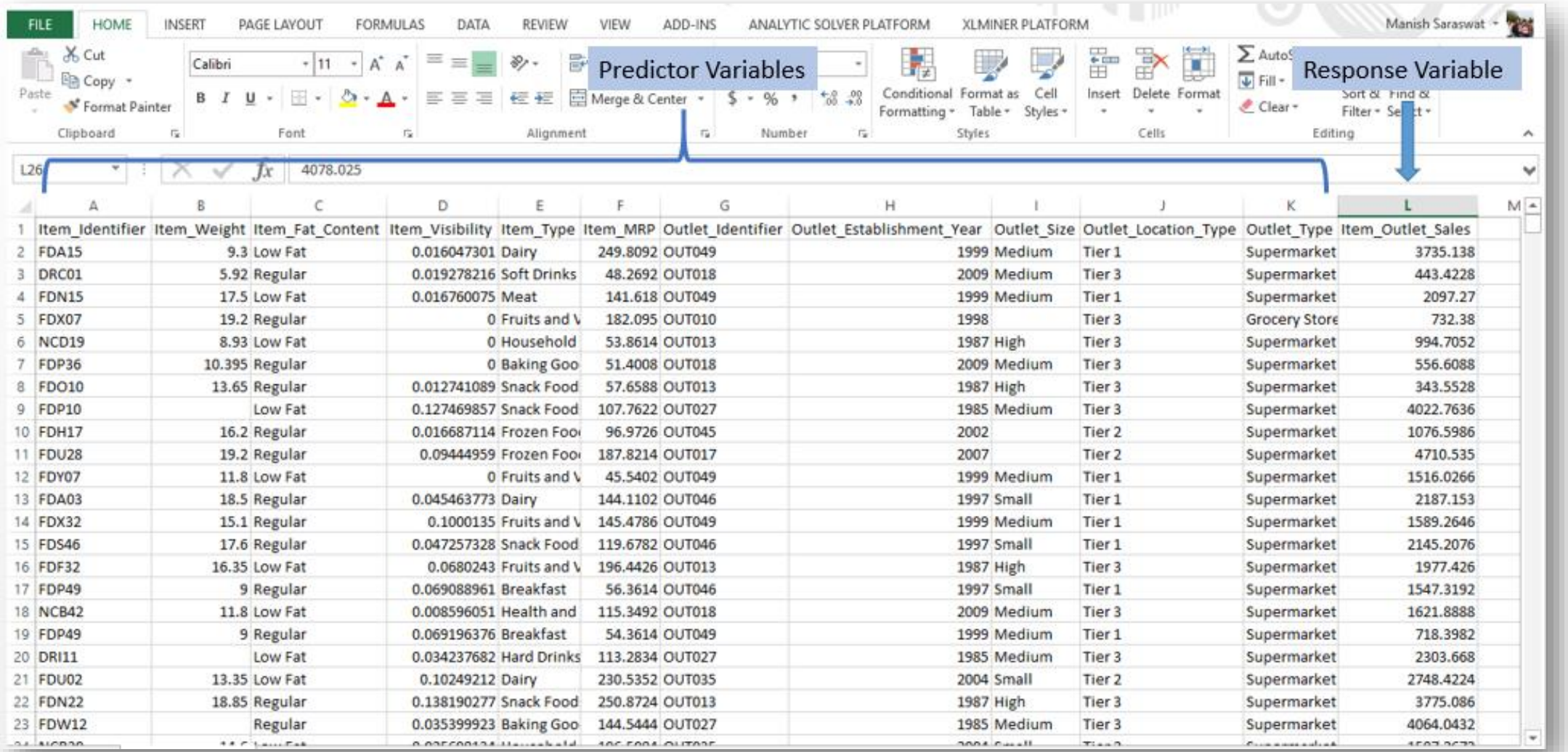
Big Mart Sales

Given sales data for 1559 products across 10 stores of the Big Mart chain in various cities the task is to build a model to predict sales for each particular product in different stores. 빅마트체인은 여러도시에 10개의 매장을 가지고 있고 1559개의 상품을 취급하고 있습니다. 매장의 세일데이터를 이용하여 세일을 예측하는 모델을 만드시오.

Big Mart Sales Data Set 데이터셋

독립변수

종속변수



	A	B	C	D	E	F	G	H	I	J	K	L
1	Item_Identifier	Item_Weight	Item_Fat_Content	Item_Visibility	Item_Type	Item_MRP	Outlet_Identifier	Outlet_Establishment_Year	Outlet_Size	Outlet_Location_Type	Outlet_Type	Item_Outlet_Sales
2	FDA15	9.3	Low Fat	0.016047301	Dairy	249.8092	OUT049	1999	Medium	Tier 1	Supermarket	3735.138
3	DRC01	5.92	Regular	0.019278216	Soft Drinks	48.2692	OUT018	2009	Medium	Tier 3	Supermarket	443.4228
4	FDN15	17.5	Low Fat	0.016760075	Meat	141.618	OUT049	1999	Medium	Tier 1	Supermarket	2097.27
5	FDX07	19.2	Regular	0	Fruits and V	182.095	OUT010	1998		Tier 3	Grocery Store	732.38
6	NCD19	8.93	Low Fat	0	Household	53.8614	OUT013	1987	High	Tier 3	Supermarket	994.7052
7	FDP36	10.395	Regular	0	Baking Goo	51.4008	OUT018	2009	Medium	Tier 3	Supermarket	556.6088
8	FDO10	13.65	Regular	0.012741089	Snack Food	57.6588	OUT013	1987	High	Tier 3	Supermarket	343.5528
9	FDP10		Low Fat	0.127469857	Snack Food	107.7622	OUT027	1985	Medium	Tier 3	Supermarket	4022.7636
10	FDH17	16.2	Regular	0.016687114	Frozen Foo	96.9726	OUT045	2002		Tier 2	Supermarket	1076.5986
11	FDU28	19.2	Regular	0.09444959	Frozen Foo	187.8214	OUT017	2007		Tier 2	Supermarket	4710.535
12	FDY07	11.8	Low Fat	0	Fruits and V	45.5402	OUT049	1999	Medium	Tier 1	Supermarket	1516.0266
13	FDA03	18.5	Regular	0.045463773	Dairy	144.1102	OUT046	1997	Small	Tier 1	Supermarket	2187.153
14	FDX32	15.1	Regular	0.1000135	Fruits and V	145.4786	OUT049	1999	Medium	Tier 1	Supermarket	1589.2646
15	FDS46	17.6	Regular	0.047257328	Snack Food	119.6782	OUT046	1997	Small	Tier 1	Supermarket	2145.2076
16	FDF32	16.35	Low Fat	0.0680243	Fruits and V	196.4426	OUT013	1987	High	Tier 3	Supermarket	1977.426
17	FDP49	9	Regular	0.069088961	Breakfast	56.3614	OUT046	1997	Small	Tier 1	Supermarket	1547.3192
18	NCB42	11.8	Low Fat	0.008596051	Health and	115.3492	OUT018	2009	Medium	Tier 3	Supermarket	1621.8888
19	FDP49	9	Regular	0.069196376	Breakfast	54.3614	OUT049	1999	Medium	Tier 1	Supermarket	718.3982
20	DRI11		Low Fat	0.034237682	Hard Drinks	113.2834	OUT027	1985	Medium	Tier 3	Supermarket	2303.668
21	FDU02	13.35	Low Fat	0.10249212	Dairy	230.5352	OUT035	2004	Small	Tier 2	Supermarket	2748.4224
22	FDN22	18.85	Regular	0.138190277	Snack Food	250.8724	OUT013	1987	High	Tier 3	Supermarket	3775.086
23	FDW12		Regular	0.035399923	Baking Goo	144.5444	OUT027	1985	Medium	Tier 3	Supermarket	4064.0432

Variables 변수들

세일에 연관된 변수들

종속변수

Variable	Description
Item_Identifier 아이템 아이디	Unique product ID
Item_Weight 아이템 무게	Weight of product
Item_Fat_Content 지방함유	Whether the product is low fat or not
Item_Visibility 아이템 노출도	% of total display area in store allocated to this product
Item_Type 아이템 종류	Category to which product belongs
Item_MRP 아이템 가격	Maximum Retail Price (list price) of product
Outlet_Identifier 아웃렛 ID	Unique store ID
Outlet_Establishment_Year 설립년도	Year in which store was established
Outlet_Size 아웃렛사이즈	Size of the store
Outlet_Location_Type 위치종류	Type of city in which store is located
Outlet_Type 아웃렛종류	Grocery store or some sort of supermarket
Item_Outlet_Sales 아웃렛세일	Sales of product in particular store. This is the outcome variable to be predicted.

Multiple Regression Result
회귀분석결과

```
Call:
lm(formula = Item_Outlet_Sales ~ ., data = new_train)

Residuals:
    Min      1Q  Median      3Q     Max
-4282.2  -672.7   -89.6   572.7  7900.6

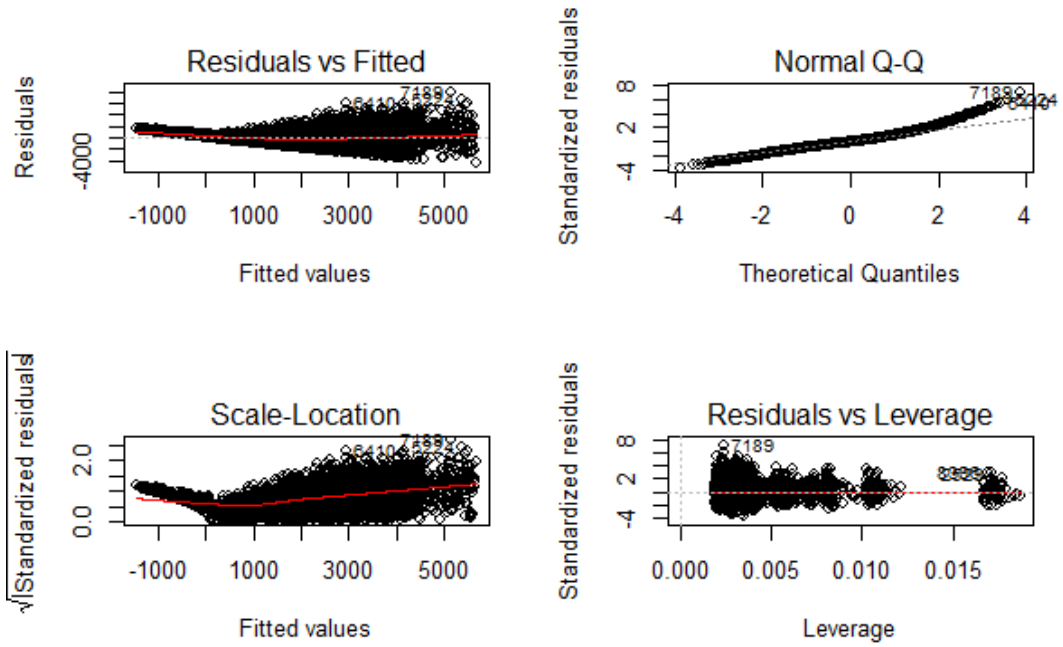
Coefficients:
                                Estimate Std. Error t value Pr(>|t|)
(Intercept)                     -980.3363   301.0963  -3.256  0.00113 **
Item_Weight                       -0.7469     2.8948  -0.258  0.79641
Item_Visibility                 -213.0599   262.2362  -0.812  0.41654
Item_MRP                          15.5618     0.1964  79.233  < 2e-16 ***
Outlet_SizeHigh                  841.6870   256.0732   3.287  0.00102 **
Outlet_SizeMedium                172.8904    71.0209   2.434  0.01494 *
Outlet_SizeSmall                 143.9449    45.6464   3.153  0.00162 **
Outlet_Location_TypeTier 2       -96.8821    90.1298  -1.075  0.28244
Outlet_Location_TypeTier 3      -321.9949   154.7523  -2.081  0.03749 *
Outlet_TypeSupermarket Type1    1478.5020   140.6138  10.515  < 2e-16 ***
Outlet_TypeSupermarket Type2    1081.5123   135.9721   7.954 2.04e-15 ***
Outlet_TypeSupermarket Type3    3635.9980   177.9708  20.430  < 2e-16 ***
Year                             -34.4801    10.4822  -3.289  0.00101 **
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 1129 on 8510 degrees of freedom
Multiple R-squared:  0.5632,    Adjusted R-squared:  0.5626
F-statistic: 914.6 on 12 and 8510 DF,  p-value: < 2.2e-16
```

Measures the goodness of fit of a regression model. Higher the r^2, better is the model. r squared가 높아야 좋은 모델

Residual Plot 잔차그래프

- Residuals = actual - predicted
  잔차 = 실제값 - 예측값
- Wedge shape → heteroscedasticity (unequal variance in errors) → transformation of dependent variable.
  웻지모양 → 종속변수 변환



Regression Result after Log Transformation 로그변환 후 회귀분석 결과

```
Call:
lm(formula = log(Item_Outlet_Sales) ~ ., data = new_train)

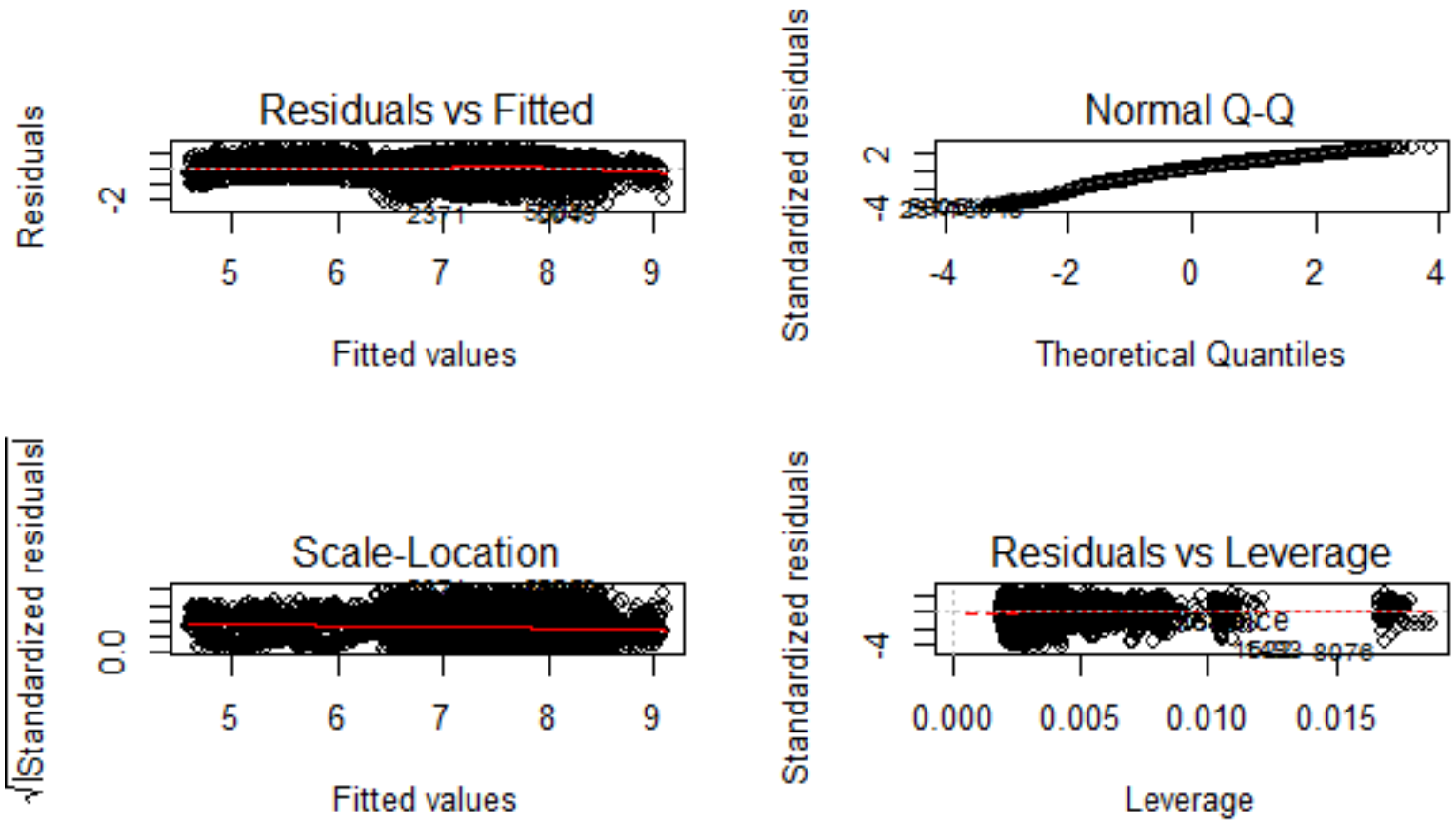
Residuals:
     Min       1Q   Median       3Q      Max
-2.28885 -0.29300  0.06825  0.38043  1.35313

Coefficients:
                                Estimate Std. Error t value Pr(>|t|)
(Intercept)                    4.751e+00  1.436e-01  33.095  < 2e-16 ***
Item_Weight                   -7.113e-04  1.380e-03  -0.515  0.60630
Item_Visibility               -4.493e-02  1.250e-01  -0.359  0.71933
Item_MRP                       8.302e-03  9.364e-05  88.656  < 2e-16 ***
Outlet_SizeHigh                3.461e-01  1.221e-01   2.835  0.00460 **
Outlet_SizeMedium              8.349e-02  3.386e-02   2.466  0.01369 *
Outlet_SizeSmall               6.761e-02  2.176e-02   3.107  0.00190 **
Outlet_Location_TypeTier 2    -3.831e-02  4.297e-02  -0.892  0.37262
Outlet_Location_TypeTier 3    -1.518e-01  7.378e-02  -2.057  0.03968 *
Outlet_TypeSupermarket Type1   1.747e+00  6.704e-02  26.057  < 2e-16 ***
Outlet_TypeSupermarket Type2   1.543e+00  6.483e-02  23.798  < 2e-16 ***
Outlet_TypeSupermarket Type3   2.602e+00  8.485e-02  30.667  < 2e-16 ***
Year                          -1.457e-02  4.998e-03  -2.916  0.00356 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.5381 on 8510 degrees of freedom
Multiple R-squared:  0.7205,    Adjusted R-squared:  0.7201
F-statistic:  1828 on 12 and 8510 DF,  p-value: < 2.2e-16
```

Improved!!

Residual Plot 잔차그래프



No more trend in residual plot! 변환 후 잔차의 추세가 사라짐

회귀분석의 종류

Linear Regression Analysis 선형회귀

- The study of how a dependent variable y is related to an independent variable. 독립변수가 한 개일 때

Multiple Regression Analysis 다중회귀

- The study of how a dependent variable y is related to two or more independent variables. 독립변수가 두개 이상 일 때

버틀러 트럭회사 예제 (선형회귀)

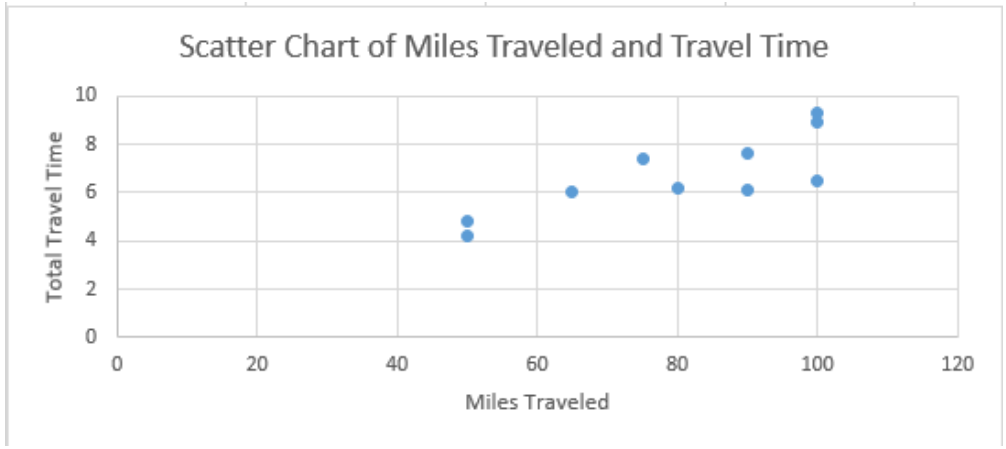
버틀러 트럭회사는 남캘리포니아에 있는 독립 트럭회사 입니다. 주요 비지니스는 로컬지역에 배달을 하는 것인데, 작업 스케쥴을 향상시키기 위해 매니저들은 각 배달원들에 대한 하루당 총 운전시간을 예측하려고 합니다. 하루당 총운전시간은 하루당 배달을 위해 움직이는 마일수와 관련이 있다고 생각해서 10개의 운전할당량을 샘플로 하여 조사를 하였습니다. 다음 슬라이드에 샘플에 대한 데이터가 제공되어 있습니다.

Miles Traveled and Travel Time Data for 10 Driving Assignments

Driving Assignment i	Miles Traveled x_i	Travel Time (hours) y_i
1	100	9.3
2	50	4.8
3	100	8.9
4	100	6.5
5	50	4.2
6	80	6.2
7	75	7.4
8	65	6
9	90	7.6
10	90	6.1

```
butler = pd.read_excel('butler.xlsx')
```

산점도 (Scatter Plot)



- Assumes data falls in a straight line! 관계가 직선이라고 가정

```
y = butler.TravelTime

x = butler.MilesTraveled

plt.grid()

plt.title('Scatter Chart of
Miles Traveled and Travel Time')

plt.xlabel('Miles Traveled')

plt.ylabel('Total Travel Time')
```

선형 회귀식 (Linear Regression Equation)

$$\hat{y}_i = b_0 + b_1 x_i$$

- $\hat{y}_i$ is the predicted value of the dependent variable 종속변수의 예측값
- b_0 is the y-intercept y절편
- b_1 is the slope 기울기
- x_i is the independent variable 독립변수

최소제곱법 (Least Squares Method)

Driving Assignment	x_i	y_i	$x_i - \bar{x}$	$y_i - \bar{y}$	$(xi - \bar{x})\ ^*(y_i - \bar{y})$	$(xi - \bar{x})^2$	Predicted Sales
1	100	9.3	20	2.6	52	400	8.0565
2	50	4.8	-30	-1.9	57	900	4.6652
3	100	8.9	20	2.2	44	400	8.0565
4	100	6.5	20	-0.2	-4	400	8.0565
5	50	4.2	-30	-2.5	75	900	4.6652
6	80	6.2	0	-0.5	0	0	6.7000
7	75	7.4	-5	0.7	-3.5	25	6.3609
8	65	6	-15	-0.7	10.5	225	5.6826
9	90	7.6	10	0.9	9	100	7.3783
10	90	6.1	10	-0.6	-6	100	7.3783
AVG	$\bar{x}$ = 80	$\bar{y}$ = 6.7		TOTAL	234	3450	

기울기, Y절편, 추정회귀식 (Slope, Y Intercept, and Estimated Regression Equation)

- Slope

$$b_1 = \frac{\sum(x_i - \bar{x})(y_i - \bar{y})}{\sum(x_i - \bar{x})^2}$$

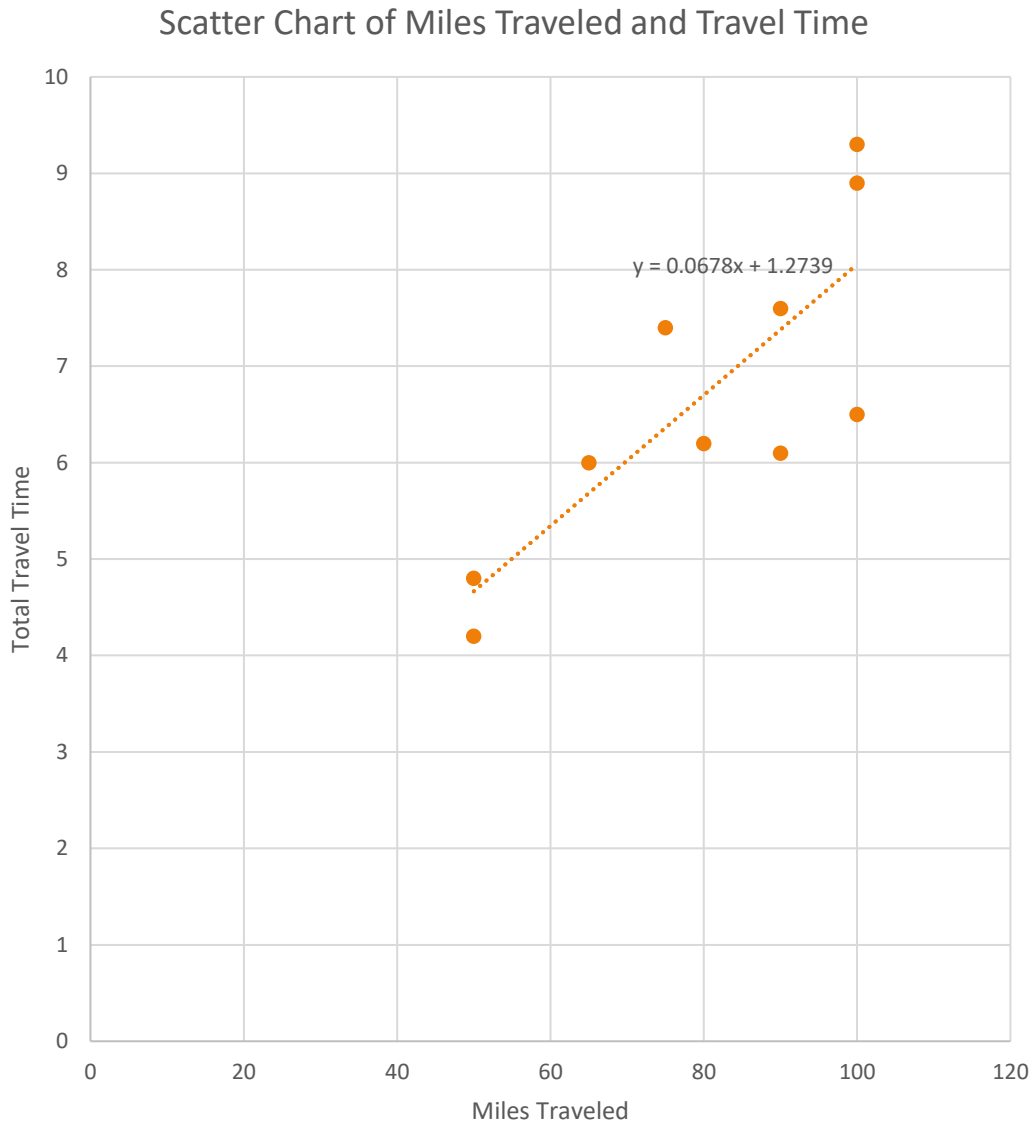
$$= \frac{234}{3450} = 0.0678$$

- Y-Intercept

$$b_0 = \bar{y} - b_1\bar{x}$$

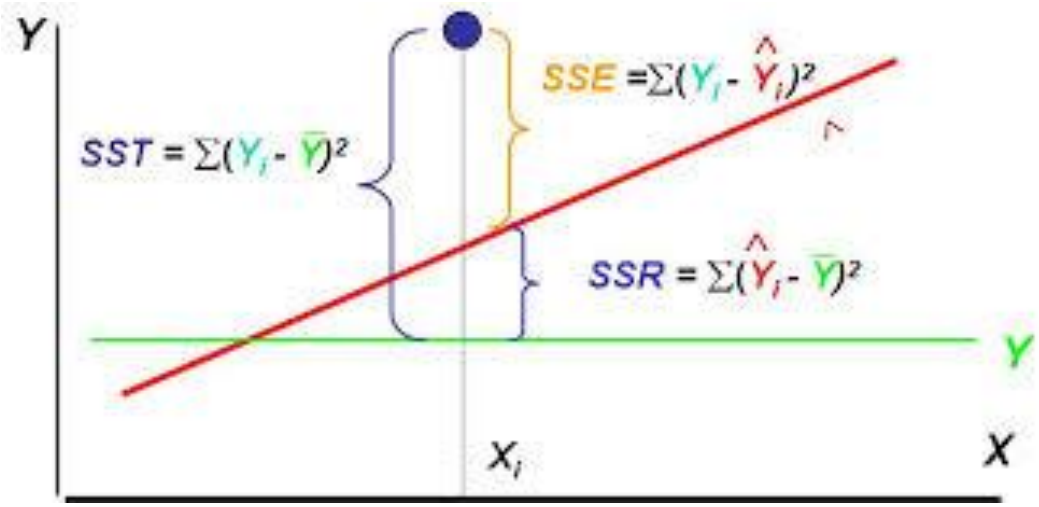
$$= 6.7 - 0.0678 * 80 = 1.2739$$

$\hat{y}_i$ = 1.2739 + 0.0678 x



결정계수 (Coefficient of Determination)

- A measure of the goodness of fit for the estimated regression equation. 회귀식의 적합성을 보여줌
- $r^2 = \frac{SSR}{SST}$



SST = SSR + SSE

Where

- SST = total sum of squares
- SSR = sum of squares due to regression
- SSE = sum of squares due to error

Sum of Squares Due to Error (오차로 인한 제곱의 합)

Driving Assignment	x_i	y_i	Predicted Sales $\hat{y}_i$	Error $y_i - \hat{y}_i$	Squared Error $(y_i - \hat{y}_i)^2$
1	100	9.3	8.0565	1.2435	1.5462
2	50	4.8	4.6652	0.1348	0.0182
3	100	8.9	8.0565	0.8435	0.7115
4	100	6.5	8.0565	-1.5565	2.4228
5	50	4.2	4.6652	-0.4652	0.2164
6	80	6.2	6.7000	-0.5000	0.2500
7	75	7.4	6.3609	1.0391	1.0798
8	65	6	5.6826	0.3174	0.1007
9	90	7.6	7.3783	0.2217	0.0492
10	90	6.1	7.3783	-1.2783	1.6340
AVG	80	6.7		SSE	8.0287

- SSE = $\sum(y_i - \hat{y}_i)^2 = 8.0287$

Total Sum of Squares (제곱의 총합)

Driving Assignment	x_i	y_i	Deviation $y_i - \bar{y}$	Squared Deviation $(y_i - \bar{y})^2$
1	100	9.3	2.6	6.76
2	50	4.8	-1.9	3.61
3	100	8.9	2.2	4.84
4	100	6.5	-0.2	0.04
5	50	4.2	-2.5	6.25
6	80	6.2	-0.5	0.25
7	75	7.4	0.7	0.49
8	65	6	-0.7	0.49
9	90	7.6	0.9	0.81
10	90	6.1	-0.6	0.36
AVG	80	$\bar{y}$ = 6.7	SST	23.9

- SST = $\sum(y_i - \bar{y})^2 = 23.9$

R Squared (결정계수)

- SSR = SST - SSE = 23.9 - 8.0287 = 15.8713
- R Squared = 15.8713 / 23.9 = .6641
- 66.41% of the variability in travel time can be explained by the linear effect of the number of miles traveled. 운전 시간 변동성의 66.41%는 이동 마일 수의 선형 효과로 설명할 수 있음 (=운전시간의 66.41%에 대한 설명력을 가짐)
- A second independent variable can be added to explain some of remaining variability in the dependent variable. 종속 변수에 남아 있는 변동성 일부를 설명하기 위해 두 번째 독립 변수를 고려해야 함

회귀분석 결과

	A	B	C	D	E	F	G	H	I
1	Driving Assignment	Miles Traveled	Travel Time (hours)						
2	1	100	9.3			Linear Model			
3	2	50	4.8			y = 1.2739 + 0.0678 x			
4	3	100	8.9						
5	4	100	6.5						
6	5	50	4.2						
7	6	80	6.2						
8	7	75	7.4						
9	8	65	6						
10	9	90	7.6						
11	10	90	6.1						
12		x	y						
13	SUMMARY OUTPUT								
14									
15	*Regression Statistics*								
16	Multiple R	0.814905707							
17	R Square	0.664071312							
18	Adjusted R Square	0.622080226							
19	Standard Error	1.001791873							
20		10							
21									
22									
23		*df*	*SS*	*MS*	*F*	*Significance F*			
24	Regression	1	15.87130435	15.87130435	15.81457814	0.004080177			
25	Residual	8	8.028695652	1.003586957					
26	Total	9	23.9						
27									
28		*Coefficients*	*Standard Error*	*t Stat*	*P-value*	*Lower 95%*	*Upper 95%*	*Lower 95.0%*	*Upper 95.0%*
29	Intercept	1.273913043	1.400744525	0.909454237	0.389687361	-1.956209623	4.50403571	-1.956209623	4.50403571
30	Miles Traveled	0.067826087	0.017055637	3.976754725	0.004080177	0.028495716	0.107156458	0.028495716	0.107156458

Slope

Intercept

```
import statsmodels.api as sm

x = sm.add_constant(x)

model = sm.OLS(y, x).fit()

model.summary()
```

버틀러 트럭회사 (다중회귀 예제)

매니저들은 또다른 독립변수를 찾기로 했는데, 배달 갯수가 총운전시간에 영향을 미칠거라고 생각했습니다. 다음 슬라이드에서 배달의 갯수를 합한 데이터 테이블을 보여줍니다. 회귀분석 명령어를 이용하여 운전한 마일수와 배달의 갯수를 독립변수로 하여 다중회귀방정식을 구하십시오.

Data For Butler Trucking with Two Independent Variables (독립변수 2개인 데이터)

Driving Assignment	Miles Traveled	Number of Deliveries	Travel Time (hours)
1	100	4	9.3
2	50	3	4.8
3	100	4	8.9
4	100	2	6.5
5	50	2	4.2
6	80	2	6.2
7	75	3	7.4
8	65	4	6
9	90	3	7.6
10	90	2	6.1

다중 회귀식 (Multiple Regression Equation)

	A	B	C	D	E	F	G	H	I
1	Driving Assignment	Miles Traveled	Number of Deliveries	Travel Time (hours)					
2	1	100	4	9.3		Linear Model			
3	2	50	3	4.8		y = -.8687 + .0611 x1 + .9234 x2			
4	3	100	4	8.9					
5	4	100	2	6.5					
6	5	50	2	4.2					
7	6	80	2	6.2					
8	7	75	3	7.4					
9	8	65	4	8					
10	9	90	3	7.5					
11	10	90	2	6.1					
12			x	y					
13	SUMMARY OUTPUT								
14									
15	*Regression Statistics*								
16	Multiple R	0.950678166							
17	R Square	0.903788975							
18	Adjusted R Square	0.876300111							
19	Standard Error	0.573142152							
20	Observations	10							
21									
22	ANOVA								
23		*df*	*SS*	*MS*	*F*	*Significance F*			
24	Regression	2	21.60055651	10.80027826	32.87836743	0.00027624			
25	Residual	7	2.299443486	0.328491927					
26	Total	9	23.9						
27									
28		*Coefficients*	*Standard Error*	*t Stat*	*P-value*	*Lower 95%*	*Upper 95%*	*Lower 95.0%*	*Upper 95.0%*
29	Intercept	-0.868701467	0.951547725	-0.912935257	0.391634304	-3.118754293	1.38135136	-3.118754293	1.38135136
30	Miles Traveled	0.061134599	0.009888495	6.182396959	0.000452961	0.037752025	0.084517173	0.037752025	0.084517173
31	Number of Deliveri	0.923425367	0.221113461	4.176251251	0.004156622	0.400575115	1.446275618	0.400575115	1.446275618

Slope for Miles Traveled

Intercept

Slope for Number of Deliveries

```
y = butler.TravelTime

x = butler.drop(['TravelTime',
'DrivingAssignment'], axis=1)

x = sm.add_constant(x)

model = sm.OLS(y,x).fit()

model.summary()
```

Travel Time = -.8687
+ .0611 Miles Traveled
+ .9234 Number of Deliveries

Interpreting the Coefficients (계수의 해석)

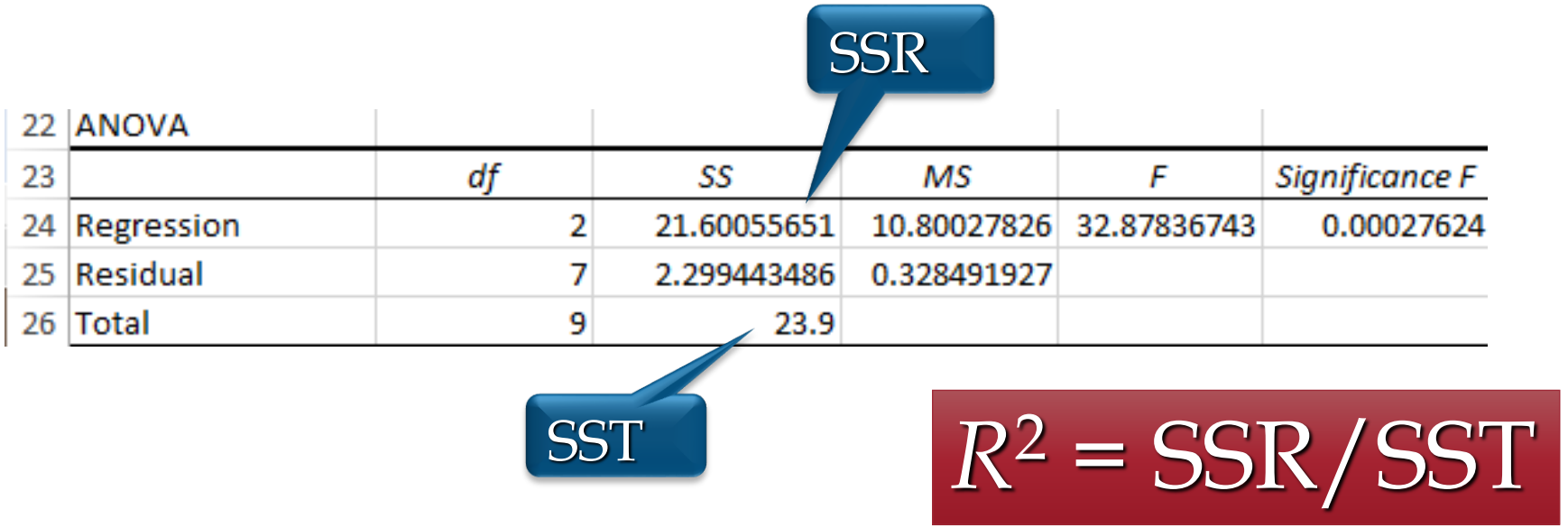
b1 = .0611

- Travel Time is expected to increase by .6011 hours for each additional Miles Traveled (when the variable number of deliveries is held constant) 운전거리가 하나 늘어날때 운전시간은 .0611 증가함

$b_2 = 0.9234$

- Travel Time is expected to increase by .9234 hours for each additional Number of Deliveries (when the variable Miles Traveled is held constant) 배달수가 하나 늘어날때 운전시간은 .9234 늘어남

Multiple Coefficient of Determination (다중결정계수)



22	ANOVA					
23		*df*	*SS*	*MS*	*F*	*Significance F*
24	Regression	2	21.60055651	10.80027826	32.87836743	0.00027624
25	Residual	7	2.299443486	0.328491927		
26	Total	9	23.9			

$R^2 = SSR/SST$

- R Squared = 21.6006/23.9 = .9038; 90.38% of the variability in Travel Time y is explained by the miles traveled and the number of deliveries as the independent variables. 운전시간 y의 변동성의 90.38%는 이동거리와 배달개수로 설명됨

Adjusted Multiple Coefficient of Determination (조정다중결정계수)

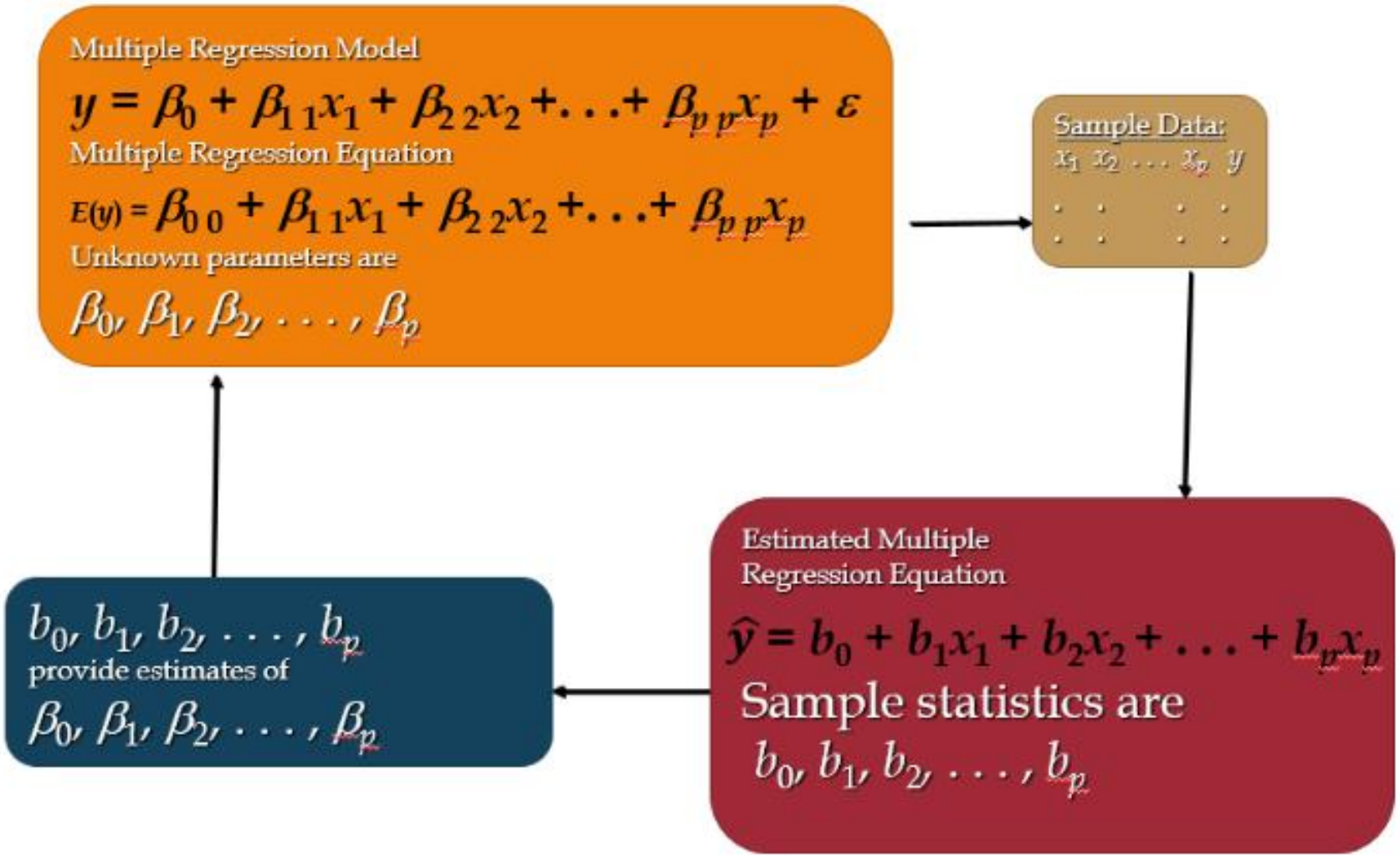
$$R_a{}^2 = 1 - (1 - R^2)\frac{n-1}{n-p-1}$$

- Where
  - n denotes the number of observations
  - p denotes the number of independent variables

$$\mathrm{R_a}^2 = 1 - (1 - .9038)\frac{10-1}{10-2-1} = .8863$$

- The adjusted multiple coefficient of determination compensates for the number of independent variables in the model. 다중결정계수는 독립변수의 갯수에 의해 조정됨

Estimation Process for Multiple Regression 다중 회귀를 위한 추정 과정



Assumptions About the Error Term ε 오차에 관한 가정

- The error ε is a random variable with mean of zero. 평균이 0임
- The variance of ε, denoted by σ2, is the same for all values of the independent variables. 독립변수에 대해서 오차가 퍼져 있는 정도가 똑같음
- The values of ε are independent. 오차끼리 독립적임
- The error ε is a normally distributed random variable reflecting the deviation between the y value and the expected value of y given by β0 + β1x1 + β2x2 + . . + βpxp. 오차는 정규분포를 따름

Testing for Significance (유의성검정)

F test for overall significance

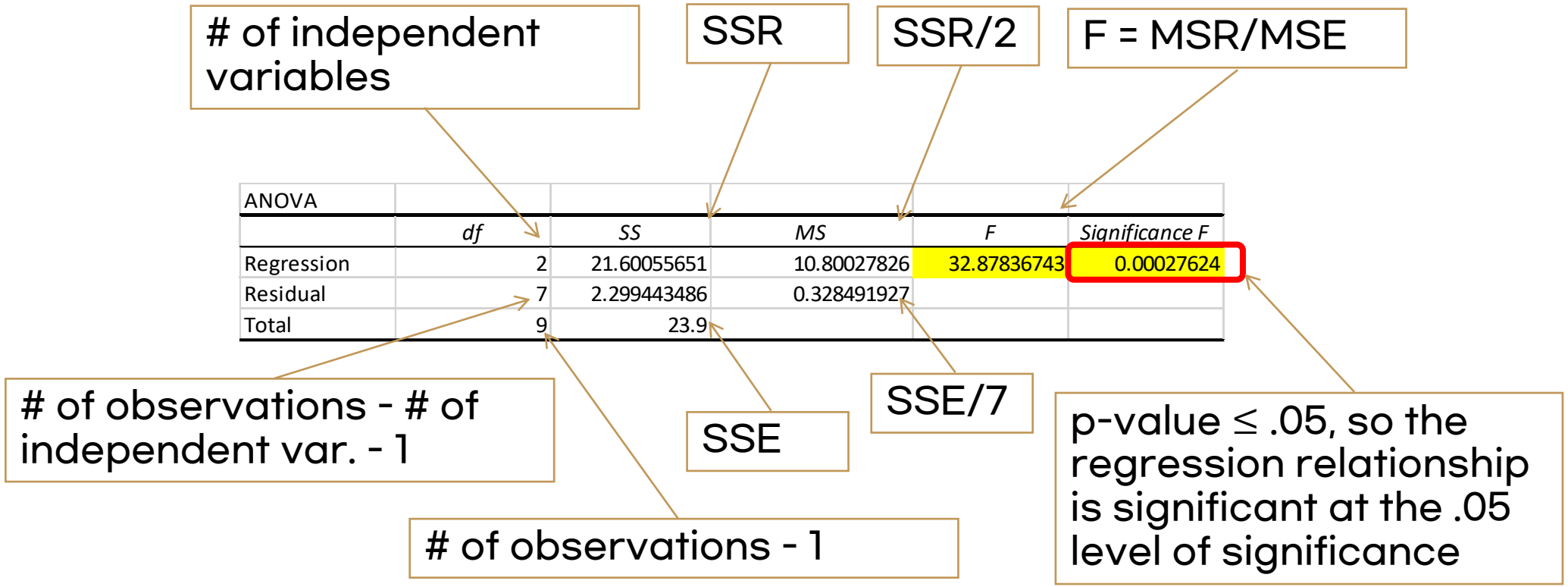
	A	B	C	D	E	F	G	H	I
25	ANOVA								
26		*df*	*SS*	*MS*	*F*	*Significance F*			
27	Regression	2	21.60055651	10.80027826	32.87836743	0.00027624			
28	Residual	7	2.299443486	0.328491927					
29	Total	9	23.9						
30									
31		*Coefficients*	*Standard Error*	*t Stat*	*P-value*	*Lower 95%*	*Upper 95%*	*Lower 95.0%*	*Upper 95.0%*
32	Intercept	-0.868701467	0.951547725	-0.912935257	0.391634304	-3.118754293	1.38135136	-3.118754293	1.38135136
33	Miles Traveled	0.061134599	0.009888495	6.182396959	0.000452961	0.037752025	0.084517173	0.037752025	0.084517173
34	Number of Deliveries	0.923425367	0.221113461	4.176251251	0.004156622	0.40057511			275618

t test for individual significance

- In simple linear regression, the F and t tests provide the same conclusion. In multiple regression, the F and t tests have different purposes. 단순회귀에서는 F, t값의 결과가 같으나 다중회귀에서는 F, t값의 결과가 다름

F Test (F 검정)

p값이 0.05보다 작기때문에
전체모델은 통계적으로 유의함
(95%의 유의성수준)



F Test Hypothesis (F 검정가설)

H_0: $\beta_1 = \beta_2 = \ldots = \beta_p = 0$
H_a: One or more of the parameters is not equal to zero

- For α = .05 and d.f. = 2, 7; F.05 = 4.74
- F = MSR/MSE = 10.8003/.3285 = 32.9
- p-value ≤ .05 (Also, F = 32.9 ≥ 4.74), so we can reject H0
  기울기가 같다는 가정을 거절함. 기울기가 있다는 것은 그 변수가 의미가 있다는 것!!
- If H0 is rejected, the test gives us sufficient statistical evidence to conclude that one or more of the parameters is not equal to zero and that the overall relationship between travel time and the miles traveled and number of deliveries is significant. p값이 0.05보다 작으므로 전체모델(두개의 변수)은 통계적으로 유의함

T Test (T 검정)

P값이 0.05보다 작으므로 운전거리는 통계적으로 유의한 변수임

T = coefficient / standard error

p-value ≤ .05, so the miles traveled is statistically significant

	Coefficients	*Standard Error*	*t Stat*	*P-value*	*Lower 95%*	*Upper 95%*	*Lower 95.0%*	*Upper 95.0%*
Intercept	-0.868701467	0.951547725	-0.912935257	0.391634304	-3.118754293	1.38135136	-3.118754293	1.38135136
Miles Traveled	0.061134599	0.009888495	6.182396959	0.000452961	0.037752025	0.084517173	0.037752025	0.084517173
Number of Deliveries	0.923425367	0.221113461	4.176251251	0.004156622	0.400575115	1.446275618	0.400575115	1.446275618

p-value ≤ .05, so the number of deliveries is statistically significant

P값이 0.05보다 작으므로 배달수는 통계적으로 유의한 변수임

T Test Hypothesis (T 검정가설)

H_0: $\beta_i = 0$
H_a: $\beta_i \neq 0$

- For α = .05 and d.f. = 7; t.025 = 2.365
- Miles traveled: t = $\frac{b_i}{s_{b_i}}$ = .0611/.0099 = 6.1717
- p-value ≤ .05 (Also, t = 6.1717 >= 2.365), so we can reject H0 .
- If H0 is rejected, the test gives us sufficient statistical evidence to conclude that the parameter is not equal to zero and that the relationship between travel time and miles traveled is significant.
  t테스트의 p값이 0.05보다 작으면 운전거리가 통계적으로 중요한 요소가 됨 (귀무가설의 거절, 다시말해 기울기가 0과 같지 않으므로 그 변수가 의미를 가짐)

Steps for Regression Modeling
회귀모델링 단계

Frequency table, histogram, descriptive statistics 기술통계 및 일변량 분석

Scatter plot, correlation matrix 다변량 분석

Regression analysis (stepwise, forward, backward) → Prediction equation 회귀분석

Residual analysis → outlier, transformation 잔차분석 및 이상치, 변환등



Residual Analysis (잔차분석)

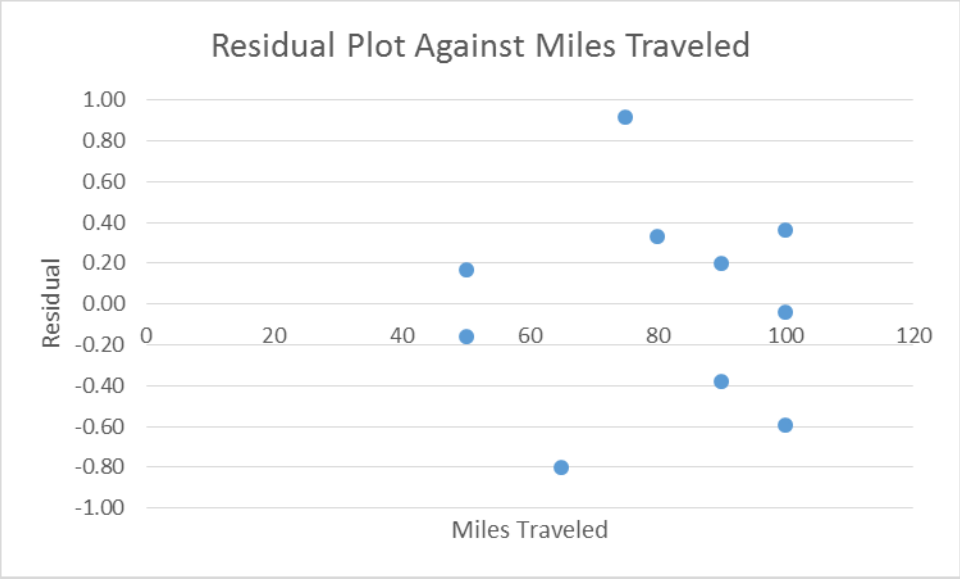
```
y_pred = model.predict(x)
residual = y - y_pred
```

Travel Time (hours)	Predicted Travel Time	Residual
9.3	8.9385	0.3615
4.8	4.9583	-0.1583
8.9	8.9385	-0.0385
6.5	7.0916	-0.5916
4.2	4.0349	0.1651
6.2	5.8689	0.3311
7.4	6.4867	0.9133
6	6.7987	-0.7987
7.6	7.4037	0.1963
6.1	6.4803	-0.3803

Residual = travel time - predicted travel time

- Residual - The difference between the observed value of the dependent variable and the predicted value of dependent variable. 종속변수와 종속변수의 예측값의 차이
- The residuals provide the best information about error; hence an analysis of the residuals is an important step in determining whether the assumptions for ε are appropriate. 잔차는 오차에 대한 중요한 정보를 주기때문에 오차가 가정에 위배되지 않는지 확인하기 위해 잔차분석을 함

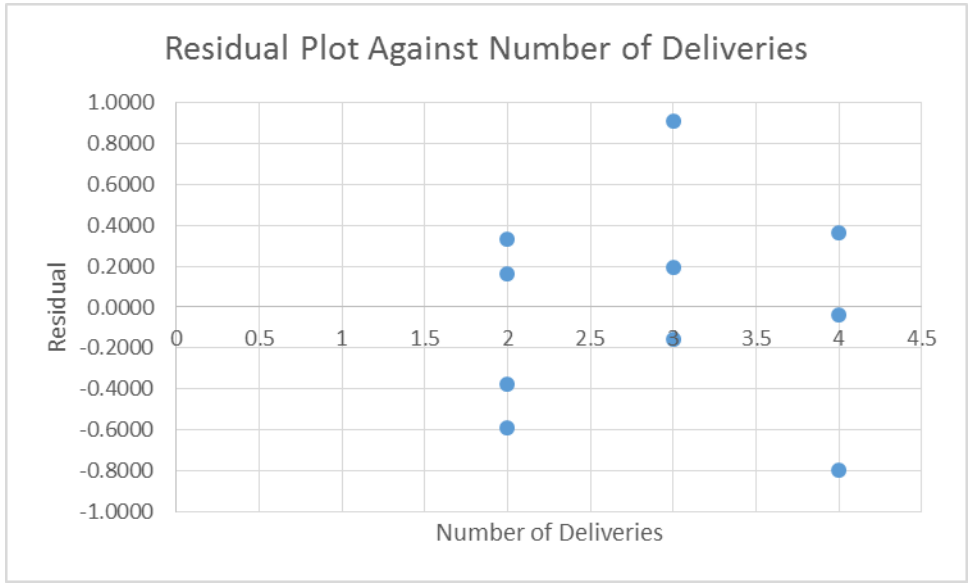
Residual Plot Against x1 (x1에 대한 잔차도 분석)



- A graph in which the values of the independent variable are represented by the horizontal axis and the corresponding residual values are represented by the vertical axis. x에는 독립변수값, y에는 잔차값

```
plt.scatter(butler.MilesTraveled,
residual)
plt.xlabel('Miles Traveled')
plt.ylabel('Residual')
plt.title('Residual Plot Against
Miles Traveled')
plt.grid()
```

Residual Plot Against x2 (x2에 대한 잔차도분석)




```
plt.scatter(butler.NumberOfDeliveries, residual)

plt.xlabel('Number Of Deliveries')

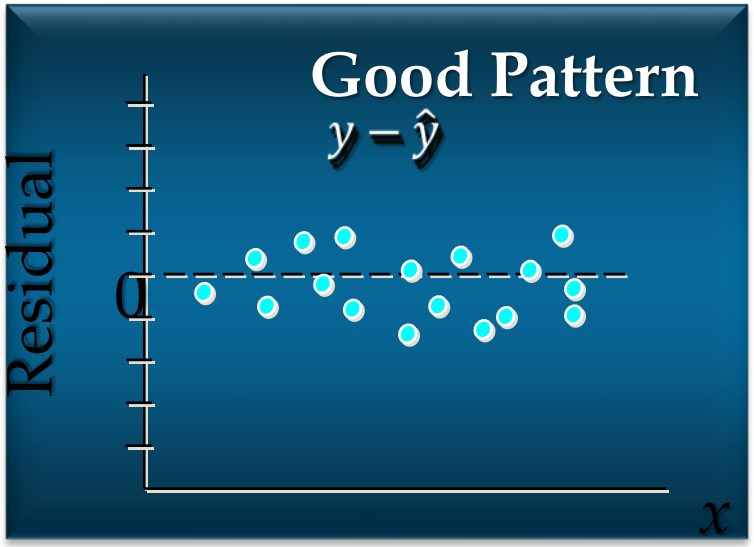
plt.ylabel('Residual')

plt.title('Residual Plot Against Number Of Deliveries')

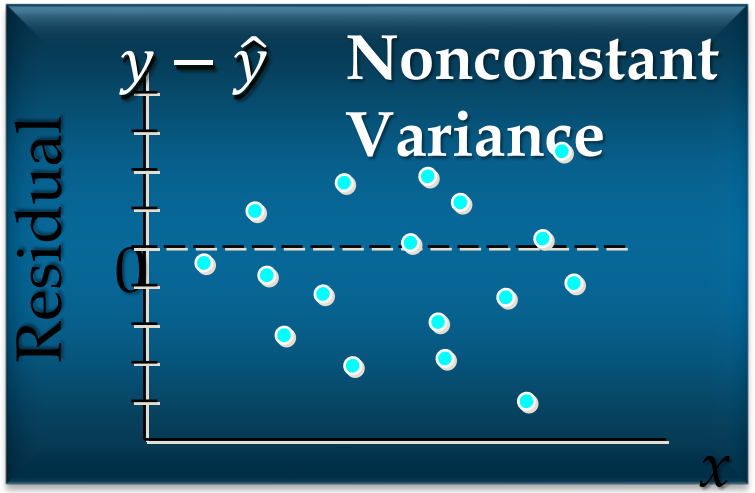
plt.grid()
```

The horizontal pattern of the residuals indicates that regression model is valid.
수평적 패턴을 보이면 회귀모델이 타당하다고 봄

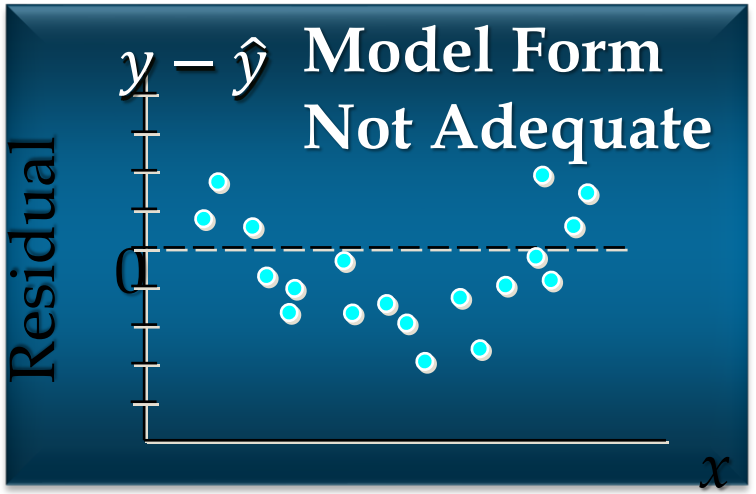
Pattern of Residual Plot 잔차도분석의 패턴



좋은 패턴



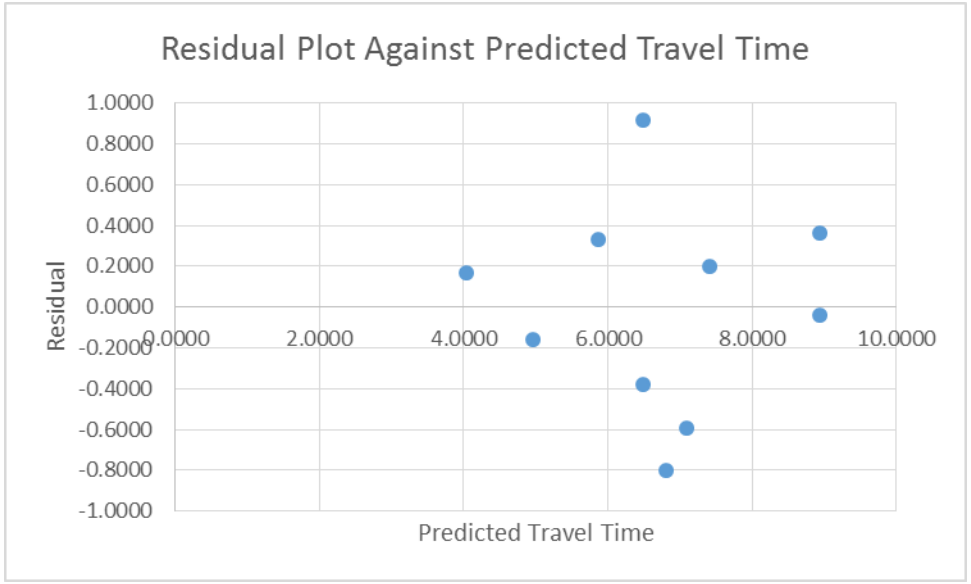
분산이 일정치 않음



모델이 적절하지 않음

Residual Plot Against $\hat{y}$ ($\hat{y}$에 대한 잔차도분석)

- The predicted value of the dependent variable $\hat{y}$ on the horizontal axis and the residual values on the vertical axis
  x축에 y의 예측값, y축에 잔차표시
- For multiple regression analysis, it is more widely used because of the presence of more than one independent variable 다중회귀분석시 독립변수가 한개이상이므로 주로 사용



수평적 패턴을 보여서 회귀모델이 타당하다고 봄

```
plt.scatter(y_pred, residual)
plt.xlabel('Predicted y')
plt.ylabel('Residual')
plt.title('Residual Plot Against
Predicted y')
plt.grid()
```

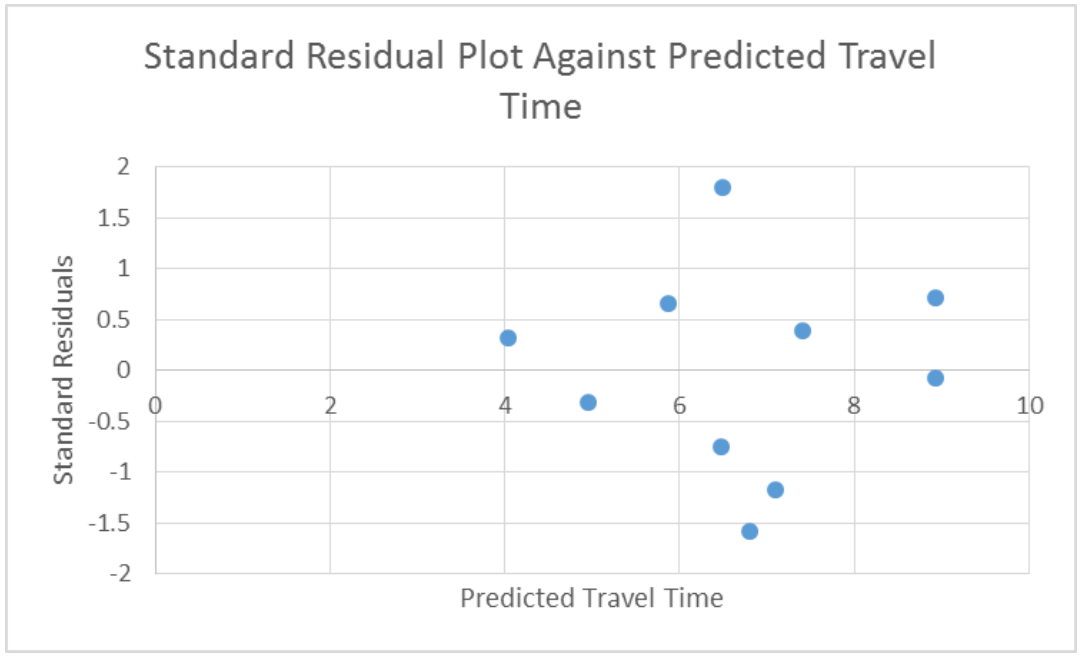
Standard Residuals 표준잔차

- Standard Residual = $\frac{y_i - \hat{y}_i}{s_{y_i - \hat{y}_i}}$
- Standard Deviation $(s_{y_i - \hat{y}_i}) = s\sqrt{1 - h_i}$
- h_i = $\frac{1}{n} + \frac{(x_i - \bar{x})^2}{\sum(x_i - \bar{x})^2}$

```
std_residual = residual / np.std(residual)
plt.scatter(y_pred, std_residual)
plt.xlabel('Predicted y')
plt.ylabel('Standard Residual')
plt.title('Standard Residual Plot Against
Predicted y')
plt.grid()
```

RESIDUAL OUTPUT			
Observation	*Predicted Travel Time (hours)*	*Residuals*	*Standard Residuals*
1	8.938459879	0.361540121	0.715264027
2	4.958304573	-0.158304573	-0.313186725
3	8.938459879	-0.038459879	-0.076088286
4	7.091609146	-0.591609146	-1.170428164
5	4.034879206	0.165120794	0.326671805
6	5.86891717	0.33108283	0.655007908
7	6.486669543	0.913330457	1.806915424
8	6.798748921	-0.798748921	-1.580229516
9	7.403688525	0.196311475	0.38837885
10	6.480263158	-0.380263158	-0.752305324

Standard Residual Plot 표준잔차도 ($\hat{y}$에 대한 표준잔차도 분석)



	A	B	C	D
5	RESIDUAL OUTPUT			
6				
7	*Observation*	*Predicted Travel Time (hours)*	*Residuals*	*Standard Residuals*
8	1	8.938459879	0.361540121	0.715264027
9	2	4.958304573	-0.158304573	-0.313186725
0	3	8.938459879	-0.038459879	-0.076088286
1	4	7.091609146	-0.591609146	-1.170428164
2	5	4.034879206	0.165120794	0.326671805
3	6	5.86891717	0.33108283	0.655007908
4	7	6.486669543	0.913330457	1.806915424
5	8	6.798748921	-0.798748921	-1.580229516
6	9	7.403688525	0.196311475	0.38837885
7	10	6.480263158	-0.380263158	-0.752305324
		x		y

```
import seaborn as sns
sns.residplot(y_pred, std_residual)
```

- Typically, standard residuals < -2 or > +2 are outliers. Providing insight about the assumption that the error term ε has a normal distribution. 2이하 또는 2이상은 이상치임! 오차가 정규분포를 띠는지 보여줌

크레이븐스 예제

크레이븐스는 여러 지역에 상품을 판매하는 회사로 각 지역에 세일대표 한명씩이 배정되어 있습니다. 크레이븐스의 데이터셋은 8개의 변수를 가진 25개의 관측치로 구성되어 있습니다. 각지역의 판매를 잘 설명해 줄수 있는 독립변수를 결정하기 위해 회귀분석을 수행하십시오. 25개 세일지역의 랜덤샘플과 변수들에 대한 정의가 다음 슬라이드에 정리되어 있습니다.

Cravens Data

Sales	Time	Poten*	AdvExp	Share	Change	Accounts*	Work	Rating
3669.88	43.1	74065.1	4582.9	2.51	0.34	74.86	15.05	4.9
3473.95	108.13	58117.3	5539.8	5.51	0.15	107.32	19.97	5.1
2295.1	13.82	21118.5	2950.4	10.91	-0.72	96.75	17.34	2.9
4675.56	186.18	68521.3	2243.1	8.27	0.17	195.12	13.4	3.4
6125.96	161.79	57805.1	7747.1	9.15	0.5	180.44	17.64	4.6
2134.94	8.94	37806.9	402.4	5.51	0.15	104.88	16.22	4.5
5031.66	365.04	50935.3	3140.6	8.54	0.55	256.1	18.8	4.6
3367.45	220.32	35602.1	2086.2	7.07	-0.49	126.83	19.86	2.3
6519.45	127.64	46176.8	8846.2	12.54	1.24	203.25	17.42	4.9
4876.37	105.69	42053.2	5673.1	8.85	0.31	119.51	21.41	2.8
2468.27	57.72	36829.7	2761.8	5.38	0.37	116.26	16.32	3.1
2533.31	23.58	33612.7	1991.8	5.43	-0.65	142.28	14.51	4.2
2408.11	13.82	21412.8	1971.5	8.48	0.64	89.43	19.35	4.3
2337.38	13.82	20416.9	1737.4	7.8	1.01	84.55	20.02	4.2
4586.95	86.99	36272	10694.2	10.34	0.11	119.51	15.26	5.5
2729.24	165.85	23093.3	8618.6	5.15	0.04	80.49	15.87	3.6
3289.4	116.26	26878.6	7747.9	6.64	0.68	136.58	7.81	3.4
2800.78	42.28	39572	4565.8	5.45	0.66	78.86	16	4.2
3264.2	52.84	51866.1	6022.7	6.31	-0.1	136.58	17.44	3.6
3453.62	165.04	58749.8	3721.1	6.35	-0.03	138.21	17.98	3.1
1741.45	10.57	23990.8	861	7.37	-1.63	75.61	20.99	1.6
2035.75	13.82	25694.9	3571.5	8.39	-0.43	102.44	21.66	3.4
1578	8.13	23736.3	2845.5	5.15	0.04	76.42	21.46	2.7
4167.44	58.44	34314.3	5060.1	12.88	0.22	136.58	24.78	2.8
2799.97	21.14	22809.5	3552	9.14	-0.74	88.62	24.96	3.9

Variable Definition

Variable	Definition
Sales	Total sales credited to the sales representative (총판매량)
Time	Length of time employed in months (고용기간, 달)
Poten	Market potential; total industry sales in units for the sales territory* (시장잠재력)
AdvExp	Advertising expenditure in the sales territory (광고비용)
Share	Market share; weighted average for the past four years (시장점유율)
Change	Change in the market share over the previous four years (시장점유율의 변화)
Accounts	Number of accounts assigned to the sales representative* (계좌수)
Work	Workload; a weighted index based on annual purchases and concentrations of accounts (작업량)
Rating	Sales representative overall rating on eight performance dimensions; an aggregate rating on a 1-7 scale (판매담당자의 전체등급)

* These data were coded to preserve confidentiality.

Lab #4 매출 예측 실습

- Descriptive statistics 기술통계
  - Summary statistics for all variables (describe) 모든 변수에 대한 기초통계
  - Summary statistics (mean, median, mode, variance, standard deviation, coefficient of variance, and skewness) for Sales 총판매량에 대한 통계

Descriptive Statistics

- Summary statistics for all variables (describe). 모든 변수에 대한 기초통계

Index	Sales	Time	Poten	AdvExp	Share	Change	Accounts	Work	Rating
count	25	25	25	25	25	25	25	25	25
mean	3374.57	87.638	38858.1	4357.39	7.5648	0.0956	122.699	18.0608	3.744
std	1313.07	86.7983	15714.5	2689.1	2.45979	0.621048	45.578	3.67007	0.959201
min	1578	8.13	20416.9	402.4	2.51	-1.63	74.86	7.81	1.6
25%	2408.11	13.82	23990.8	2243.1	5.51	-0.1	88.62	16	3.1
50%	3264.2	57.72	36272	3571.5	7.37	0.15	116.26	17.64	3.6
75%	4167.44	127.64	50935.3	5673.1	8.85	0.5	136.58	20.02	4.5
max	6519.45	365.04	74065.1	10694.2	12.88	1.24	256.1	24.96	5.5

```
cravens.describe()
```

Statistics for Sales

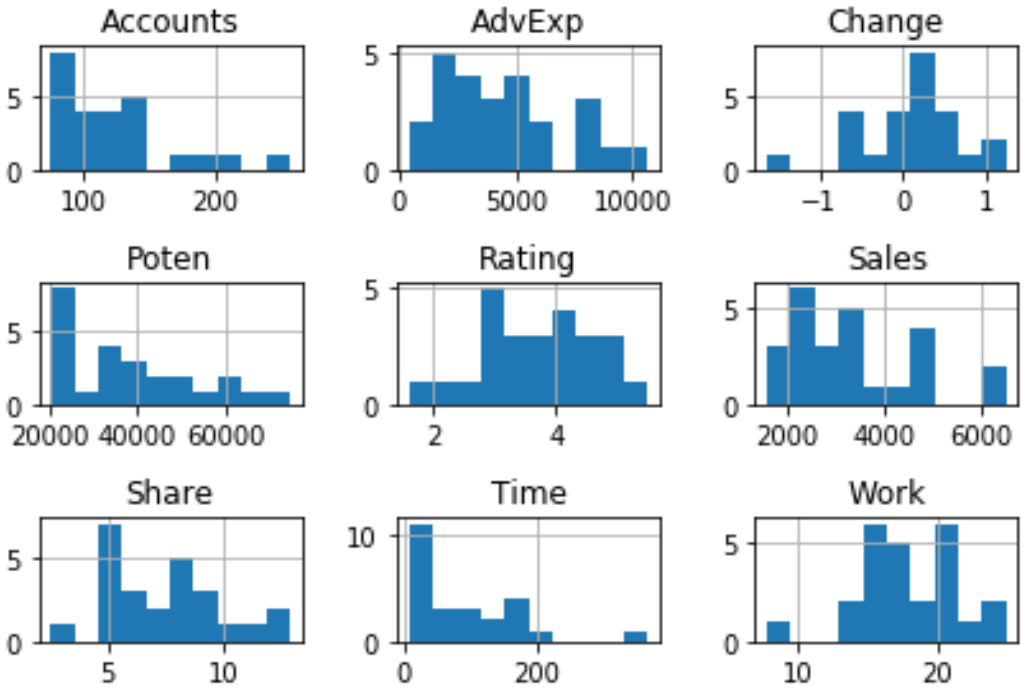
- Summary statistics (mean, median, mode, variance, standard deviation, coefficient of variance, and skewness) for Sales 총판매량에 대한 통계

```
sales = cravens.Sales
sales.mean() #평균
sales.median() #중위수
sales.mode() #최빈수
sales.var() #분산
sales.std() #표준편차
(sales.std()/sales.mean
  ())*100 #변동계수
sales.skew() #기움
```

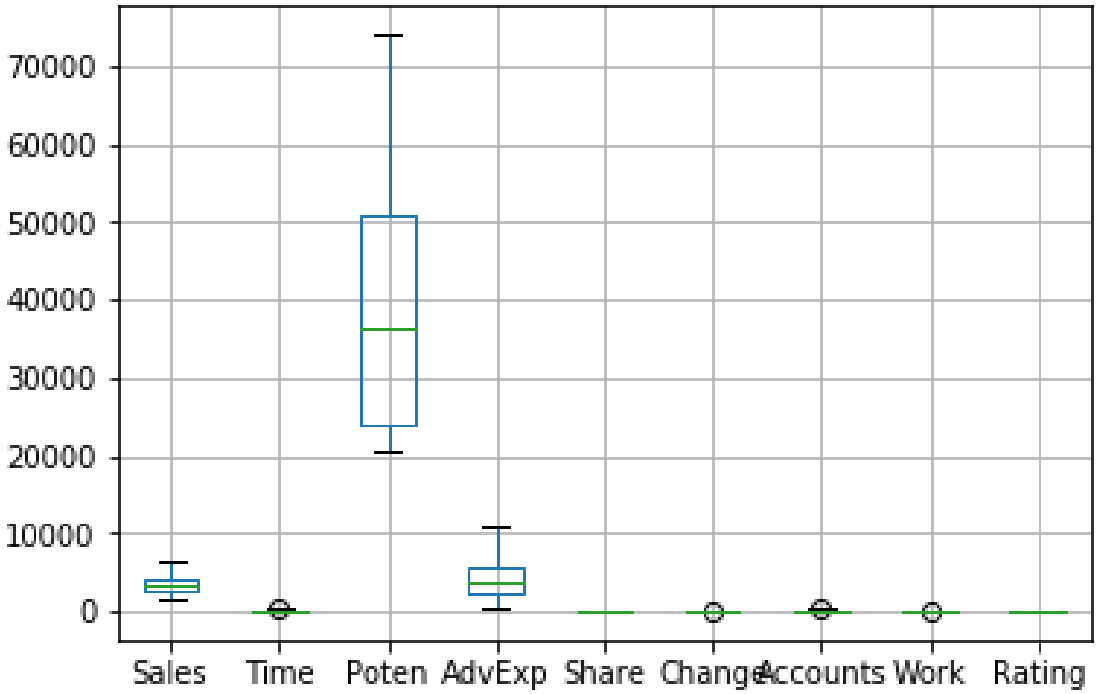
Lab #4 매출 예측 실습

- Univariate analysis 일변량분석
  - Frequency, percent frequency, histogram, and boxplot. 빈도표, 백분율빈도표, 히스토그램, 박스플랏
- Multivariate analysis 다변량분석
  - Correlation matrix, scatter plot (pairplot). 상관관계, 산점도

Histogram and Boxplot




```
cravens.hist()
plt.tight_layout()
```




```
cravens.boxplot()
```

Frequency Table for Sales

Index	freq	r_freq	p_freq
(1499.999, 2000.0]	2	0.08	8
(2000.0, 2500.0]	6	0.24	24
(2500.0, 3000.0]	4	0.16	16
(3000.0, 3500.0]	5	0.2	20
(3500.0, 4000.0]	1	0.04	4
(4000.0, 4500.0]	1	0.04	4
(4500.0, 5000.0]	3	0.12	12
(5000.0, 5500.0]	1	0.04	4
(5500.0, 6000.0]	0	0	0
(6000.0, 6500.0]	1	0.04	4
(6500.0, 7000.0]	1	0.04	4

```
(sales.max() - sales.min())/9 #class
    9 width 500

breaks = np.arange(1500, 7500, 500)

sales_cut = pd.cut(sales, breaks,
    include_lowest=True)

freq =
    sales_cut.value_counts(sort=False)

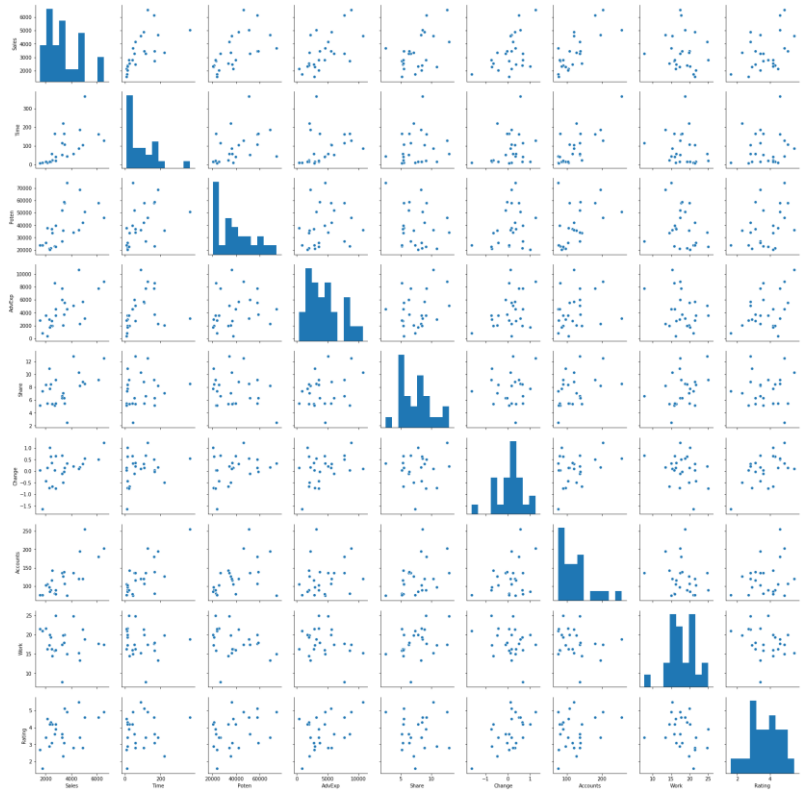
r_freq = freq.values/len(sales)

p_freq = r_freq *100

freq_table = pd.DataFrame({
        'freq':freq.values,
        'r_freq':r_freq,
        'p_freq':p_freq
        }, index=freq.index)

sales.hist(bin=breaks, rwidth=.9)
```

Scatter Plot



```
from pandas.plotting
   import scatter_matrix
scatter_matrix(cravens)
```

```
import seaborn as sns
sns.pairplot(cravens)
```

Correlation Matrix

Index	Sales	Time	Poten	AdvExp	Share	Change	Accounts	Work	Rating
Sales	1	0.622881	0.597812	0.596177	0.483511	0.489181	0.753986	-0.117223	0.401878
Time	0.622881	1	0.453978	0.24917	0.106105	0.251469	0.757789	-0.179406	0.101172
Poten	0.597812	0.453978	1	0.1741	-0.210671	0.268287	0.478643	-0.258839	0.358703
AdvExp	0.596177	0.24917	0.1741	1	0.264458	0.376516	0.200037	-0.272235	0.411462
Share	0.483511	0.106105	-0.210671	0.264458	1	0.0854699	0.403011	0.349345	-0.0235629
Change	0.489181	0.251469	0.268287	0.376516	0.0854699	1	0.327415	-0.287665	0.549333
Accounts	0.753986	0.757789	0.478643	0.200037	0.403011	0.327415	1	-0.198848	0.228613
Work	-0.117223	-0.179406	-0.258839	-0.272235	0.349345	-0.287665	-0.198848	1	-0.276913
Rating	0.401878	0.101172	0.358703	0.411462	-0.0235629	0.549333	0.228613	-0.276913	1

```
cravens.corr()
np.corrcoef(cravens)
```

Correlation Coefficient Between Independent Variables (상관계수)

Avoid including both Time and Accounts in the same regression model! 시간과 어카운트중 하나를 모델에 써야함

	Sales	*Time*	*Poten*	*AdvExp*	*Share*	*Change*	*Accounts*	*Work*	*Rating*
Sales	1								
Time	0.622881	1							
Poten	0.597812	0.453978	1						
AdvExp	0.596177	0.24917	0.1741	1					
Share	0.483511	0.106105	-0.21067	0.264458	1				
Change	0.489181	0.251469	0.268287	0.376516	0.08547	1			
Accounts	0.753986	0.757789	0.478643	0.200037	0.403011	0.327415	1		
Work	-0.11722	-0.17241	-0.25884	-0.27223	0.349345	-0.28766	-0.19885	1	
Rating	0.401878	0.[illegible]72	0.358703	0.411462	-0.02356	0.549333	0.228613	-0.27691	1

Multicolinearity

Need further consideration! 이것도 고려대상

- When the independent variables are highly correlated (say, |r | > .7), it is not possible to determine the separate effect of any particular independent variable on the dependent variable. 독립변수들 끼리의 상관관계가 높으면 종속변수에 대한 효과를 판별하기 어려움

Multicollinearity (다중공선성)

- The correlation among the independent variables. 독립변수 사이의 상관관계
- Every attempt should be made to avoid including independent variables that are highly correlated. 독립변수 사이에 상관관계가 높은 것은 피해야 함
- If the estimated regression equation is to be used only for predictive purposes, multicollinearity is usually not a serious problem. 예측모델을 개발하는 것이면 큰 문제는 안됨

다중공선성 진단법

- 결정계수 r2값은 높아 회귀식의 설명력은 높지만 식안의 독립변수의 P-value값이 커서 개별 인자들이 유의하지 않는 경우가 있다. 이런 경우 독립변수들 간에 높은 상관관계가 있다고 의심된다.
- 독립변수들간의 상관계수를 구한다.

다중공선성 해결법

- 상관관계가 높은 독립변수중 하나 혹은 일부를 제거한다.
- 변수를 변형시키거나 새로운 관측치를 이용한다.
- 자료를 수집하는 현장의 상황을 보아 상관관계의 이유를 파악하여 해결한다.
- PCA(Principle Component Analysis)를 이용한 diagnol matrix의 형태로 공선성을 없애준다.

Correlation Coefficients between Sales and Independent Variables

	Sales
Sales	1
Time	0.622881
Poten	0.597812
AdvExp	0.596177
Share	0.483511
Change	0.489181
Accounts	0.753986
Work	-0.11722
Rating	0.401878

The single best predictor of Sales!

- Give us a quick indication of which independent variables are good indicators 쉽게 중요한 독립변수를 가릴 수 있음
- R Squared (coefficient of determination) of one independent variable = sample correlation coefficient^2. 상관계수의 제곱은 결정계수와 같음
- (.754)^2 * 100 = 56.85% of the variability in Sales is explained by Accounts. 세일의 56.85%가 어카운트에 의해서 설명됨

Lab #4 매출 예측 실습

- Multiple regression with: 회귀분석
  - all variables 모든 변수사용
  - 'Poten', 'AdvExp', 'Share‘
  - 'Poten', 'AdvExp', 'Share', 'Accounts‘
  - 'Share', 'Change', 'Accounts', 'Work', 'Rating‘
  - 'Poten', 'AdvExp', 'Share', 'Change','Time‘
- Compare above models and determine a best model. 위 모델들을 비교하고 가장 좋은 모델을 선택
- Standard residual plot for the best model. 표준잔차플랏

Regression Analysis with Eight Independent Variables (모든 변수)

SUMMARY OUTPUT	(Eight-Variable Multiple Regression)							
Regression Statistics								
Multiple R	0.960230559							
R Square	0.922042726							
Adjusted R Square	0.88306409							
Standard Error	449.0153681							
Observations	25							
ANOVA								
	df	*SS*	*MS*	*F*	*Significance F*			
Regression	8	38153712.11	4769214.014	23.65507887	1.81494E-07			
Residual	16	3225836.813	201614.8008					
Total	24	41379548.93						
	Coefficients	*Standard Error*	*t Stat*	*P-value*	*Lower 95%*	*Upper 95%*	*Lower 95.0%*	*Upper 95.0%*
Intercept	-1507.835778	778.6084146	-1.936577809	0.070662983	-3158.411882	142.7403263	-3158.411882	142.7403263
Time	2.010136903	1.930514447	1.041244165	0.313241339	-2.082370903	6.102644709	-2.082370903	6.102644709
Poten	0.037205572	0.00820216	4.536070133	0.000337431	0.01981777	0.054593374	0.01981777	0.054593374
AdvExp	0.150983659	0.047106902	3.205128193	0.005517914	0.051121488	0.250845829	0.051121488	0.250845829
Share	199.0401821	67.02914902	2.969457095	0.009036923	56.94473392	341.1356303	56.94473392	341.1356303
Change	290.8665748	186.7768726	1.557294385	0.138958171	-105.0827071	686.8158567	-105.0827071	686.8158567
Accounts	5.549744788	4.775443082	1.16214238	0.262220228	-4.573742307	15.67323188	-4.573742307	15.67323188
Work	19.79389359	33.67517144	0.587788948	0.56487845	-51.59428079	91.18206797	-51.59428079	91.18206797
Rating	8.189034709	128.4984664	0.063728657	0.949975686	-264.2155452	280.5936146	-264.2155452	280.5936146

```
import statsmodels.api as sm

x = cravens.drop(['Sales'], axis=1)

y = cravens.Sales

x = sm.add_constant(x)

model = sm.OLS(y,x).fit()

model.summary()
```

- Only Poten, AdvExp, and Share are significant at the $\alpha = .05$ level, given the effect of all other variables.

Regression Analysis with Three Independent Variables (독립변수 3개)

SUMMARY OUTPUT	**(Three-Variable Multiple Regression)**
Regression Statistics	
Multiple R	0.921398731
R Square	0.848975622
Adjusted R Square	0.827400711
Standard Error	545.5150751
Observations	25

It is almost high as the eight independent variables estimated regression equation! 모든 변수를 썼을때와 비슷한 결과

ANOVA

	df	*SS*	*MS*	*F*	*Significance F*
Regression	3	35130228.29	11710076.1	39.35013294	8.43042E-09
Residual	21	6249320.641	297586.6972		
Total	24	41379548.93			

	Coefficients	*Standard Error*	*t Stat*	*P-value*	*Lower 95%*	*Upper 95%*	*Lower 95.0%*	*Upper 95.0%*
Intercept	-1603.582149	505.5509129	-3.171949863	0.00459264	-2654.932826	-552.231471	-2654.932826	-552.231471
Poten	0.054286049	0.007474124	7.263198867	3.73883E-07	0.038742757	0.06982934	0.038742757	0.06982934
AdvExp	0.167480081	0.044273275	3.782870884	0.001090565	0.075408767	0.259551396	0.075408767	0.259551396
Share	282.7469265	48.75560594	5.799270075	9.33297E-06	181.3540934	384.1397596	181.3540934	384.1397596

Regression Analysis with Four Independent Variables (독립변수 4개)

SUMMARY OUTPUT								
Regression Statistics								
Multiple R	0.948920248							
R Square	0.900449637							
Adjusted R Square	0.880539564							
Standard Error	453.8363767							
Observations	25							
ANOVA								
	df	*SS*	*MS*	*F*	*Significance F*			
Regression	4	37260199.79	9315049.948	45.22583369	9.56365E-10			
Residual	20	4119349.136	205967.4568					
Total	24	41379548.93						
	Coefficients	*Standard Error*	*t Stat*	*P-value*	*Lower 95%*	*Upper 95%*	*Lower 95.0%*	*Upper 95.0%*
Intercept	-1441.932281	423.5818169	-3.40414112	0.002814351	-2325.508468	-558.3560936	-2325.508468	-558.3560936
Poten	0.038217461	0.007976948	4.790987882	0.000111303	0.021577839	0.054857083	0.021577839	0.054857083
AdvExp	0.174990175	0.036906717	4.741418073	0.00012484	0.098004113	0.251976238	0.098004113	0.251976238
Share	190.1441924	49.74418396	3.822440681	0.00106525	86.37964295	293.9087418	86.37964295	293.9087418
Accounts	9.213922434	2.8652133	3.215789356	0.004337001	3.237192221	15.19065265	3.237192221	15.19065265

R Square is increased to .9004.

- Estimated regression equation

Sales = -1441.9323 + .03282 Poten + .1750 AdvExp + 190.1441 Share + 9.2139 Accounts

Variable Selection 변수선택법

- Compute all possible regressions. 모든 가능한 회귀분석을 다 계산
  - 255 different estimated regression equations involving one or more independent variables would have to be fitted to the data. 255개의 회귀식을 다 테스트해야함
- Select the subset of independent variables that provide the best estimated regression equation. 조합을 만들어서 테스트
- Stepwise, Forward, Backward, and Best-subsets regression. 스텝와이즈기법 이용
- Software Packages are available to conduct different types of regressions. 소프트웨어를 이용하여 해결

Variable Selection Procedures (변수선택법)

- Stepwise Regression
- Forward Selection
- Backward Elimination

} Iterative; a single independent variable at a time is added or deleted to evaluate the new model. 한개의 변수를 더하거나 빼면서 가장 좋은 모델을 찾아냄

- Best-Subsets Regression

} Different subsets of the independent variables are evaluated. 독립변수들의 부분집합을 만들어서 테스트

The first 3 procedures are heuristics and therefore offer no guarantee that the best model will be found. 휴리스틱한 방법이므로 가장 좋은 모델이 아닐 수도 있음